Unnamed_Unit

Unit 3 - Designing Solutions

PDF Documents

PDF Document 1: 1738067081038-Chapter 5_unit 3 - signing Solutions.pdf

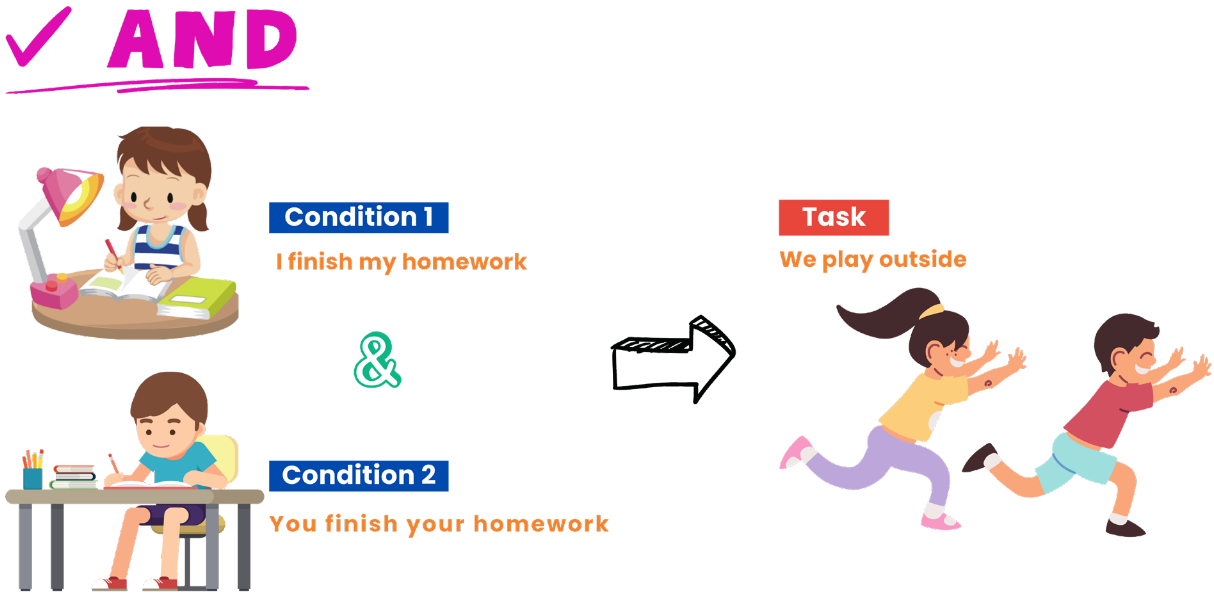
This PDF document is included in the unit materials.

UNIT 3 DESIGNING SOLUTIONS

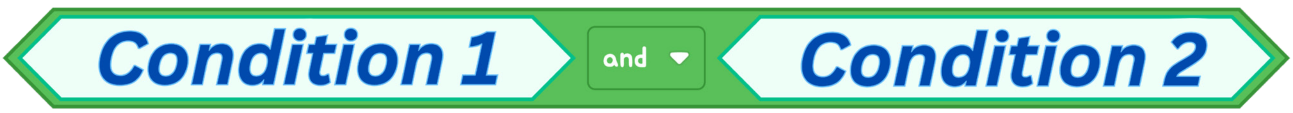
Programming Concepts

Imagine you and your friend want to play outside, but both of you have homework to finish. You agree that you can only go out if both of you complete your homework. This agreement is like using the "AND" operator in programming.

If you say, "I will finish my homework AND you will finish yours," it means both conditions must be true for you to go outside. If either of you doesn't finish your homework, you both stay inside.



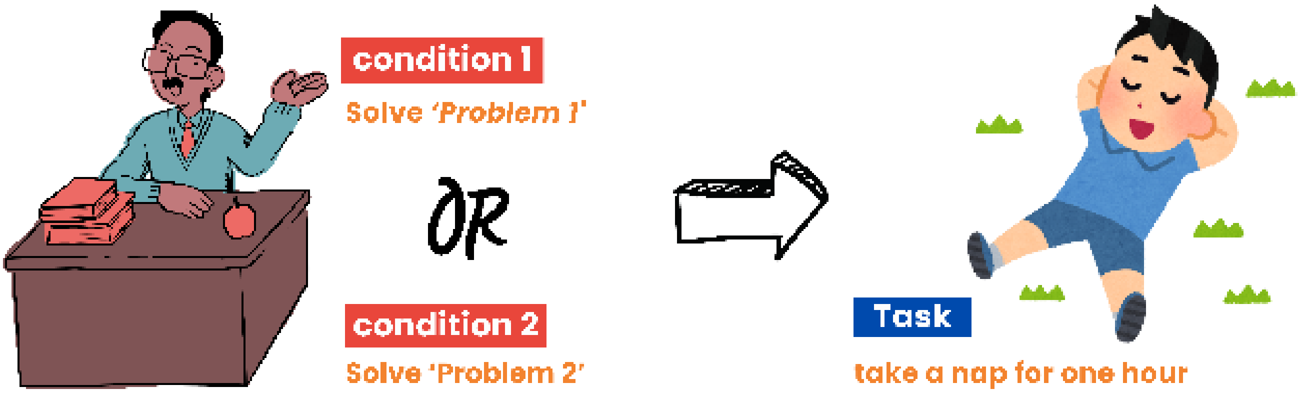
In programming, when you use the "AND" operator in an if statement, it ensures that both conditions must be true for the code block to execute. So, if your if statement includes two conditions separated by "AND," both must be satisfied for the code within the if statement to run.



OR

Think of the "or" operator in programming as providing options, similar to how choices might be given during school.

✓ OR



For example, imagine the principal announces during a break, "If anyone can solve either of these math problems (Problem 1 or Problem 2), then all students will get one hour of nap time today." This scenario does not require solving both problems—solving just one will grant the reward.

Similarly, in programming, using the "or" operator means if any of the conditions connected by "or" is true, the associated action will be taken.

TIPS

When using the "or" operator, it's crucial to manage the logical flow of your program effectively. If any connected condition is true, the overall condition is evaluated as true.

Flowcharts & Algorithms

Let's read the flowchart and create the code.

FLOWCHART